INCH-POUND

MIL-S-52159C
2 February 1994
SUPERSEDING
MIL-S-52159B
27 September 1971

MILITARY SPECIFICATION

SLING, MULTIPLE LEG,

.625 INCH BY 6-FOOT CHAIN,

WITH SPECIAL HOOKS AND RING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a two-leg chain sling equipped with special hooks and a connecting ring.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL-C-16173	- Corrosion Preventive Compound, Solvent Cutback, Cold Application.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US ARMY BELVOIR RDE CTR, ATTN SATBE TSE, 10101 GRIDLEY RD STE 104, FT BELVOIR VA 22060-5818 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5420

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: STDZN DCMNT ORDER DESK, BLDG 4D, 700 ROBBINS AVE, PHILADELPHIA PA 19111-5094.)

2.1.2 Other Government drawings. The following other Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those cited in the solicitation.

DRAWINGS

ME

DL13212E7099	- Sling, Multiple Leg.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from: USA BELVOIR RDE CTR, ATTN SATBE JBS, 10101 GRIDLEY RD, STE 104, FT BELVOIR VA 22060-5818.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3953	- Strapping, Flat Steel and Seals.
D 4675	- Selection and Use of Flat Strapping Materials.

(Application for copies should be addressed to: AMERCN SCTY & MTRLS, 1916 RACE STRET, PHILADELPHIA PA 19103.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC. (NMFTA)

National Motor Freight Classification Rules

(Application for copies should be addressed to: NATL MTR FRGHT TRFC ASS, ATTN TRFC ORDR SECT, 2200 MILL RD, ALEXANDRIA VA 22314.)

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UNIFORM CLASSIFICATION COMMITTEE (UCC)

Uniform Freight Classification Rules

(Application for copies should be addressed to: UNFRM FRGHT CLASS CMMTE, ATTN TARIFF PBLSHNG OFCR, ROOM 1106, 222 S RIVERSIDE PLZ, CHICAGO IL 60606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The multiple leg chain sling, hereinafter referred to as "sling", shall be in accordance with DL13212E7099 and as specified herein (see 6.2).

3.1.1 Drawings. The drawing forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to ensure correct fit, assembly, and operation of the sling. Any data (e.g., shop drawing, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available upon request, for inspection by the contracting officer or the designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Materials. Materials shall be as specified herein and on the drawings. Materials not specified shall be selected by the contractor, and shall be subject to all provisions of this specification.

3.3.1 Material deterioration and prevention and control. The sling shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the sling may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the sling may be

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newly fabricated from recovered materials to the maximum extent practicable, provided the sling produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the sling.

3.4 Performance. The assembled sling shall lift a load of 22,000 pounds for 5 minutes without rupture or permanent deformation in any of the chain links, hooks, connecting links, and the connecting ring. The included angle between the legs of the sling shall be 60 degrees ±5 degrees when lifting the load.

3.5 Identification marking. The sling shall be identified in accordance with MIL-STD-130.

3.6 Workmanship. All surfaces shall be rounded and free from sharp edges, slivers, burrs, cracks and dents or other irregularities which would impair the safe use or function of the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order and unless disapproved by the Government, the contractor's own facilities or any other, suitable for the performance of the inspection requirements specified herein, may be used. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance or rejection (see 6.4 and 6.5.1).

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article sling shall be examined as specified in 4.5.1, as applicable. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article sling shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and test shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The slings shall be examined as specified herein for the following defects:

Major

101. Material not as specified (see 3.3).
102. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration of the applicable storage and operating environments (see 3.3.1).
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.1.1).
105. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the sling (see 3.3.2).
106. Workmanship not as specified (see 3.6).
107. Incomplete or incorrect assembly.
108. Any part or component not in accordance with the QAP requirements as shown on the drawings.

Minor

201. Identification marking incorrect, missing or illegible (see 3.5).
202. Dimensions other than those identified by QAP data not as specified.

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4.5.2 Test. Suspend an 11,000-pound load for 5 minutes from each hook of the sling while the legs of the sling form an included angle of 60 degrees ±5 degrees. Any evidence of breakage or permanent deformation shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

105. Slings not coated with the preservative specified (see 5.1).
106. Boxes not as specified for Level A or B (see 5.2.1 or 5.2.2).
107. Blocking and bracing not adequate to prevent movement for level A or B (see 5.2.1 or 5.2.2).
108. Quantities packed together exceed limitations of the box for level A or B (see 5.2.1 or 5.2.2).
109. Strapping not as specified for level A or B (see 5.2.1 or 5.2.2).
110. Marking missing, illegible, incorrect, or incomplete (see 5.3).

5. PACKAGING

5.1 Preservation. Each complete sling shall be coated with preservative conforming to MIL-C-16173, grade 1 (see 6.6).

5.2 Packing. Packing shall be level A, B, or C as specified.

5.2.1 Level A. The slings shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, style optional or PPP-B-621, class 2, style optional, in quantities not to exceed the weight limitation of the box. The contents shall be blocked and braced to prevent free movement. Closure shall be in accordance with the appendix to the applicable box specification. Strapping shall be in accordance with ASTM D 3953, type 1 or 2, zinc-coated, size as applicable and ASTM D 4675.

5.2.2 Level B. Packing shall be as specified in 5.2.1 for level A except boxes shall be domestic type or class as applicable and strapping may be zinc-coated.

5.2.3 Level C. The slings shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in fiberboard or wooden containers complying with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

5.3 Marking. In addition to any special marking specified in the contract or purchase order (see 6.2), marking shall be in accordance with MIL-STD-129.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The sling is intended for use in the erection of fixed and floating bridges.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When first article is required for inspection and approval, and the number of units required (see 3.2).
c. Level of preservation and packing required (see 5.1 and 5.2).
d. Any special marking (see 5.3).

6.3 First article. When a first article inspection is required, the item(s) should be a production model. The first article should consist of 1 or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Quality assurance provisions (QAP). The contracting officer should require the supplier to maintain records of all QAP inspections. A suggested paragraph is as follows:

"The supplier shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, corrected action taken when applicable."

6.5 Definitions.

6.5.1 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.6 Levels of preservation. Levels of preservation have not been included as the preservation specified is the minimum acceptable for any level.

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6.7 Subject term (key word) listing.

Chain, sling
Fixed bridge sling
Floating bridge sling
Sling, chain
Sling, bridge erection
Sling, multiple leg
Sling, hook and ring

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

User activity:
Navy - MC

Preparing activity:
Army - ME

Project 5420-0176

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
The submitter of this form must complete blocks 4, 5, 6, and 7.
The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-52159C	2. DOCUMENT DATE (YYMMDD) 940202

3. DOCUMENT TITLE
SLING, MULTIPLE LEG, .625 INCH BY 6-FOOT CHAIN, WITH SPECIAL HOOKS AND RING

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) DSN	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Carolyn B. Johnson	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3468 (2) DSN 654-3468
c. ADDRESS (Include Zip Code) US ARMY BELVOIR RDE CTR ATTN SATBE TSE 10101 GRIDLEY RD STE 104 FT BELVOIR VA 22060-5818	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: DEFNS QLTY & STDZN OFC 5203 LEESBURG PIKE STE 1403 FLS CHURCH VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

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